

7_MODEL_EXPLANATION_GUIDE

MODEL EXPLANATION GUIDE

Why Predictions Happen - Detailed Rationale for All Outputs

FOUR PREDICTION GOALS EXPLAINED

1. Risk Classification: Low/Medium/High Mental Health Risk

How It Works:

The model analyzes 13 key factors simultaneously to determine overall risk level:

- **Primary Drivers:** Depression score, anxiety score, stress level
- **Protective Factors:** Sleep quality, social support, physical activity
- **Contextual Factors:** Age, gender, employment situation, work environment

Why Specific Results Occur:

HIGH RISK Classification Logic:

- 1 Input Example: Age 28, Female, Employed, On-site
- 2 Stress: 9/10, Sleep: 4.5 hours, Depression: 26, Anxiety: 18
- 3 Social Support: 35, Productivity: 55
- 4
- 5 WHY HIGH RISK:
- 6 ✓ Extremely high depression score (26/30) - severe symptoms
- 7 ✓ Very high anxiety score (18/21) - significant distress
- 8 ✓ Critical sleep deprivation (4.5 hours) - amplifies all other risks
- 9 ✓ Maximum stress level (9/10) - overwhelming pressure
- 10 ✓ Low social support (35/100) - no protective buffering
- 11 ✓ Impaired productivity (55/100) - functional impact evident
- 12 X No treatment history despite severe symptoms - concerning pattern
- 13

MEDIUM RISK Classification Logic:

- 1
- 2 Input Example: Age 42, Male, Employed, Hybrid

```
3 Stress: 6/10, Sleep: 6.5 hours, Depression: 16, Anxiety: 12
4 Social Support: 60, Productivity: 78
5
6 WHY MEDIUM RISK:
7 ✓ Moderate depression (16/30) - noticeable but manageable
8 ✓ Moderate anxiety (12/21) - some distress present
9 ✓ Reasonable stress level (6/10) - within normal range
10 ✓ Adequate sleep (6.5 hours) - provides some protection
11 ✓ Good social support (60/100) - buffering effect present
12 ✓ Maintained productivity (78/100) - functioning preserved
13 ✓ No treatment history - may indicate resilience or denial
14
```

LOW RISK Classification Logic:

```
1 Input Example: Age 58, Non-binary, Self-employed, Remote
2 Stress: 2/10, Sleep: 8.2 hours, Depression: 3, Anxiety: 1
3 Social Support: 92, Productivity: 96
4
5 WHY LOW RISK:
6 ✓ Minimal depression (3/30) - almost no symptoms
7 ✓ Minimal anxiety (1/21) - very low distress
8 ✓ Very low stress (2/10) - excellent coping
9 ✓ Excellent sleep (8.2 hours) - optimal restoration
10 ✓ Strong social support (92/100) - robust protection
11 ✓ High productivity (96/100) - optimal functioning
12 ✓ Healthy lifestyle factors present
13
```

2. Treatment Seeking: Likelihood of Seeking Mental Health Treatment

How It Works:

The model evaluates multiple factors that influence help-seeking behavior:

- **Symptom Severity:** Higher scores generally increase treatment likelihood
- **Demographic Patterns:** Gender, age, and cultural factors affect willingness
- **Access Factors:** Employment status and insurance coverage considerations
- **Stigma Indicators:** Mental health history and current treatment patterns

Why Treatment Predictions Occur:

HIGH LIKELIHOOD OF SEEKING TREATMENT:

Input Example: Age 35, Female, Student, Remote

Depression: 22, Anxiety: 16, Stress: 7

Mental Health History: Yes, Currently Seeks Treatment: Yes

WHY LIKELY TO SEEK TREATMENT:

- ✓ Significant symptoms (depression 22, anxiety 16) - motivation to improve
- ✓ Treatment history present - familiarity with mental health services
- ✓ Currently seeking treatment - active engagement pattern
- ✓ Female gender - statistically more likely to seek help
- ✓ Student status - often have campus mental health resources
- ✓ Remote work - may have more flexible schedule for appointments

LOW LIKELIHOOD OF SEEKING TREATMENT:

Input Example: Age 45, Male, Self-employed, On-site

Depression: 18, Anxiety: 12, Stress: 6

Mental Health History: No, Currently Seeks Treatment: No

WHY UNLIKELY TO SEEK TREATMENT:

- ✓ Moderate symptoms but not severe enough to overcome barriers
- ✓ No treatment history - may not recognize need or face stigma
- ✓ Male gender - statistically less likely to seek mental health help
- ✓ Self-employed status - cost and time barriers significant
- ✓ On-site work environment - less flexible scheduling
- ✓ May rely on self-coping rather than professional help

UNCERTAIN/MIXED INDICATORS:

Input Example: Age 29, Non-binary, Unemployed, Hybrid

Depression: 20, Anxiety: 14, Stress: 8

Mental Health History: No, Currently Seeks Treatment: No

WHY UNCERTAIN PREDICTION:

- ✓ Symptoms approaching treatment threshold (depression 20)
 - ✓ High stress (8/10) suggesting potential motivation
 - ✓ No treatment history creates uncertainty
 - ✓ Unemployment could be barrier or opportunity for treatment
 - ✓ Non-binary identity - unique factors not fully captured in model
-

3. Symptom Severity: Depression/Anxiety Score Levels

How It Works:

The model predicts likely depression (0-30) and anxiety (0-21) scores based on:

- **Direct Correlates:** Stress levels, sleep quality, social support
- **Demographic Factors:** Age and gender patterns
- **Lifestyle Indicators:** Physical activity, work environment
- **Historical Context:** Mental health history and treatment patterns

Why Specific Scores Are Predicted:

HIGH DEPRESSION SCORE PREDICTION (25-30):

Input Indicators Leading to High Prediction:

- ✓ Stress level: 8-10/10 → Strong correlation with depression
- ✓ Sleep hours: 3-5 hours → Severe sleep deprivation increases depression risk
- ✓ Social support: 20-40/100 → Low support amplifies depressive symptoms
- ✓ Physical activity: 0-2 days/week → Inactivity linked to depression
- ✓ Age: Younger adults (18-35) → Higher depression rates in this group
- ✓ Work environment: On-site high-stress jobs → Occupational depression risk
- ✓ Mental health history: Present → Recurrence likelihood

MODERATE DEPRESSION SCORE PREDICTION (15-24):

Input Indicators Leading to Moderate Prediction:

- ✓ Stress level: 5-7/10 → Moderate stress elevation
- ✓ Sleep hours: 5-7 hours → Suboptimal but not severely deficient
- ✓ Social support: 40-70/100 → Adequate but not strong support
- ✓ Physical activity: 3-4 days/week → Some beneficial activity
- ✓ Age: Middle adulthood (35-55) → Typical depression onset period
- ✓ Mixed work factors → Moderate occupational stress
- ✓ No clear protective or risk factors dominant

LOW DEPRESSION SCORE PREDICTION (0-14):

Input Indicators Leading to Low Prediction:

- ✓ Stress level: 1-4/10 → Minimal psychological stress
- ✓ Sleep hours: 7-9 hours → Optimal sleep duration
- ✓ Social support: 70-100/100 → Strong protective network

- ✓ Physical activity: 5-7 days/week → Regular mood-enhancing exercise
- ✓ Age: Older adults (50+) → Generally lower depression rates
- ✓ Favorable work/life balance → Reduced occupational stress
- ✓ Multiple protective factors present

ANXIETY SCORE PREDICTION PATTERNS:

Similar logic applies but with some key differences:

- **Gender effect:** Females typically show higher anxiety scores
- **Age effect:** Young adults show higher anxiety than seniors
- **Work environment:** Remote/hybrid often shows lower anxiety than on-site
- **Social factors:** Social support has stronger inverse correlation with anxiety

4. Intervention Response: Treatment Effectiveness Prediction

How It Works:

The model estimates how likely someone is to benefit from specific interventions based on:

- **Current symptom profile:** Severity and type of symptoms
- **Personal characteristics:** Age, gender, lifestyle factors
- **Treatment history:** Past response patterns
- **Support systems:** Social and environmental resources

Why Response Predictions Vary:

HIGH RESPONSE LIKELIHOOD:

Favorable Indicators:

- ✓ Younger age (18-35) - typically better treatment response
- ✓ Strong social support network present
- ✓ Good baseline functioning (high productivity scores)
- ✓ Willingness to seek treatment already demonstrated
- ✓ No severe comorbid conditions indicated
- ✓ Regular sleep patterns present

MODERATE RESPONSE LIKELIHOOD:

Mixed Indicators:

- ✓ Some protective factors present but not optimal
- ✓ Moderate symptom severity - enough motivation but not overwhelming
- ✓ Average social support levels
- ✓ Functional but not optimal baseline
- ✓ May require combination of interventions

CHALLENGED RESPONSE LIKELIHOOD:

Barrier Indicators:

- ✓ Severe symptom levels with long duration
- ✓ Limited social support systems
- ✓ Significant life stressors present
- ✓ Advanced age with established patterns
- ✓ Multiple comorbid conditions suggested
- ✓ Poor sleep and lifestyle factors



INPUT-OUTPUT MAPPING EXPLANATIONS

Key Decision Trees:

Risk Classification Decision Flow:

```
1  IF depression_score > 25 OR anxiety_score > 18:
2      IF sleep_hours < 5 OR stress_level > 8:
3          → HIGH RISK
4      ELSE:
5          → MEDIUM RISK
6  ELSE IF depression_score < 10 AND anxiety_score < 8:
7      IF sleep_hours > 7 AND social_support > 80:
8          → LOW RISK
9      ELSE:
10         → MEDIUM RISK
11 ELSE:
12     → MEDIUM RISK (balancing factors)
```

Treatment Seeking Decision Flow:

```
1  IF current_treatment = YES:
2      → HIGH LIKELIHOOD
3  ELSE IF gender = FEMALE AND depression_score > 15:
```

```
4         → MODERATE-HIGH LIKELIHOOD
5     ELSE IF gender = MALE AND stress_level > 7:
6         → MODERATE LIKELIHOOD
7     ELSE IF age < 30 AND social_support > 70:
8         → MODERATE LIKELIHOOD
9     ELSE:
10        → LOW LIKELIHOOD
```

Symptom Score Prediction Logic:

```
1  Depression Score = Base Level +
2      (Stress × 2.5) +
3      (Sleep Deficit × 3.0) +
4      (Low Social Support × 1.8) +
5      (Inactivity × 1.5) +
6      (Demographic Adjustment)
7
8  Anxiety Score = Base Level +
9      (Stress × 2.0) +
10     (Sleep Issues × 2.2) +
11     (Social Isolation × 1.6) +
12     (Gender Factor × 2.5) +
13     (Age Adjustment)
```

REAL-WORLD APPLICATION EXAMPLES

Example 1: Why a 22-year-old student got HIGH RISK

Input: Student, Remote, Stress=9, Sleep=4 hours, Depression=27, Anxiety=19

EXPLANATION:

"Your prediction shows HIGH RISK because the combination of being a student under significant academic pressure (stress=9), getting severely inadequate sleep (only 4 hours), and experiencing severe depressive symptoms (score 27 out of 30) creates a dangerous pattern. The model recognizes that students with these profiles often face academic consequences and social isolation, which amplifies the risk. The lack of treatment history despite severe symptoms is particularly concerning as it suggests the problem may be worsening without intervention."

Example 2: Why a 45-year-old professional got MEDIUM RISK but LOW treatment likelihood

Input: Employed, On-site, Stress=6, Sleep=6.5 hours, Depression=16, Male

EXPLANATION:

"You're classified as MEDIUM RISK because your depression score (16) and stress level (6) indicate moderate symptoms that are impacting your wellbeing. However, as a 45-year-old male professional, the model predicts you're less likely to seek treatment due to common barriers: workplace stigma, time constraints from on-site work, and societal expectations around masculinity and self-reliance. The model suggests you might benefit from flexible online therapy options or workplace mental health programs that address these specific barriers."

Example 3: Why scores improved after lifestyle changes

Previous Input: Sleep=4 hours, Exercise=1 day/week, Social Support=30

New Input: Sleep=7 hours, Exercise=5 days/week, Social Support=75

EXPLANATION:

"The model shows significant improvement in your predicted scores because you've addressed three key protective factors simultaneously. Better sleep (7 hours vs 4) directly impacts mood regulation and stress resilience. Increased physical activity (5 days vs 1) provides natural antidepressant effects through endorphin release. Enhanced social support (75 vs 30) creates buffering against stress and provides accountability for maintaining healthy habits. The model recognizes this combination as highly effective for symptom reduction."

This explanation guide ensures users understand not just what the predictions are, but why they occur and how specific inputs influence outcomes